SANIKA DHURI

AI Engineer

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WORK EXPERIENCE

Software Engineer - Citius Tech Healthcare Technology Mumbai *July 2025 - Present*

ClaimFlow AI: Agentic Claims Workflow

- Built a **sequential agentic workflow** in Python that takes claims as JSON input and validates them against PL/SQL rules before submission.
- Added **semantic search** to recommend fixes for flagged claims, so rejected claims could be corrected and submitted cleanly.
- Kept the workflow **PHI-safe** using synthetic data, with audit logging and human review before submission, reducing submission time by **40%** against a defined benchmark.

Backend Engineering (.NET REST APIs)

- Built **RESTful APIs** in C# and .NET using layered architecture and **dependency injection**, with async/await, pagination, filtering, **request validation**, AutoMapper DTO mapping, and **fallback handling** for resilient responses, secured with RBAC.
- Optimized SQL queries with multi-table joins and LINQ, cutting average API response time by **~67%** (45 sec to 15 sec).
- Delivered 340B regulatory compliance changes for the Craneware client across the API and data layers, with NUnit and Postman endpoint testing.

AI Intern - Predixion AI *July 2024 - May 2025*

- Developed an AI powered **voice calling agent** using large language models and **retrieval augmented generation** to automate the loan recovery process.
- **Reduced** manual intervention by **60%** through intelligent automation and machine learning-driven task routing.
- Integrated Twilio and ElevenLabs text-to-speech, **reducing latency by 50%** and improving conversational quality.
- Built system using **open-source libraries** and cloud infrastructure for cost-effective deployment.

PROJECTS

StratifyAI - Agentic Ticketing Resolution *GitHub Link*

- Architected a multi-intent enterprise support agent using **LangGraph** and **LangChain**, with modular orchestration nodes (intent splitting, routing, metadata extraction, validation, decision, execution) and **human-in-the-loop** confirmation to safely automate multi-ticket workflows, reducing estimated resolution time by up to **~70%** for general queries and significantly for finance workflows (multi-week to under a week), based on estimated manual baselines.
- Built a **RAG pipeline** with document ingestion, structured chunking, embeddings, JSON-based storage in PostgreSQL and NumPy **cosine similarity** for policy-aware responses across HR, Finance, IT and Travel.
- Added semantic and tool-level caching, structured logging, and persistent multi-turn state via LangGraph's PostgreSQL-backed checkpointer for continuity across sessions.
- Integrated **Opik** for observability and tracing of retrieval quality and multi-step agent orchestration, and deployed on **GCP Compute Engine** Linux VM with key-based **SSH** access, PostgreSQL setup, public access via firewall configuration.

Dividend Radar: Remote MCP Server for Indian Stock Portfolios *GitHub Link*

- Built and deployed a remote MCP server in Python (**FastMCP**) that connects Claude Desktop to **live portfolio data** through **Zerodha Kite Connect** and yfinance, exposing tools to fetch holdings, analyze dividend income, view dividend history and track recent announcements.
- Designed the Kite **authorization flow** so users authenticate directly with Zerodha and never share brokerage credentials with the server; tested with **MCP Inspector** and deployed on **FastMCP Cloud** as a public remote endpoint.

AI-Driven Early Detection of Colon Cancer: A Multi-Stage Deep Learning Approach *GitHub Link*

- Built a three-part system for early colon cancer detection covering histopathology image classification, GI tract video analysis, and RAG-based Q&A; **fine-tuned ViT**, VGG16, and ResNet50 via transfer learning on a 100K-image, 14GB dataset using an **NVIDIA RTX 5000 GPU**, reaching **93% accuracy** using **LIME** for explainability.

SKILLS

Languages & Backend: Python, SQL, PL/SQL, REST APIs, LINQ, AutoMapper, Dependency Injection, Role-based Auth, NUnit, FastAPI
AI/ML: PyTorch, Scikit-learn, NumPy, Transformers, NLP, GANs, LoRA, RAG, Agentic AI, Semantic Search, Transfer Learning, Computer Vision, Explainable AI (LIME)
Frameworks & Databases: LangChain, LangGraph, Hugging Face, MCP, Streamlit, PostgreSQL, MySQL, FAISS
Cloud & MLOps: AWS (EC2, S3, Lambda), GCP (Compute Engine), Azure OpenAI, Linux, MLflow, DVC
Tools: Git, GitHub, VS Code, Postman, LangFuse, Opik, Twilio, ElevenLabs

EDUCATION

Bachelor of Engineering *Nov 2021 - Aug 2025*
*Vivekanand Education Society's Institute of Technology (**9.05 CGPA**)*

CERTIFICATES

- NVIDIA Fundamentals of Deep Learning *Aug 2023*
- AWS Cloud Foundation *Mar 2024*